HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and applied advanced **structural and thermo-mechanical FEA methodologies** for server SoCs and advanced semiconductor packages, including **CPI, package assembly, warpage, stress, and deformation analysis**.
- Analyzed both **soldered and socketed packages** under various conditions including package assembly, SMT, socket loading, and thermal cycling.
- Modeled the **package-to-platform mechanical interaction** of SoC packages, PCBs, sockets, heatsinks, and **thermal interface materials (TIM1.5)**.
- Translated complex **simulation results** into clear design requirements and recommendations for cross-functional engineering teams and leadership.
- Partnered with computational modeling teams to align modeling assumptions, material properties, and boundary conditions for accurate analysis.
- Developed physics-based reliability models for failure mechanisms such as **solder fatigue and substrate cracking**, quantifying damage accumulation.
- Containerized FEA pre- and post-processing workflows using **Docker**, enabling consistent execution across development and analysis environments on **AWS EC2**.

SKILLS

FEA & Simulation – Finite Element Analysis, Structural Mechanics, Thermo-Mechanical Analysis, Package Mechanics, Chip-Package Interaction (CPI), Warpage, Stress Analysis, Deformation Analysis, Solder Joint Reliability, Creep, Fatigue, Fracture Mechanics, Damage Mechanics, Submodeling, Contact Analysis, Material Nonlinearity, Geometric Nonlinearity, Viscoelasticity, Coupled Structural-Thermal Analysis

Software & Tools – ANSYS APDL, ANSYS Workbench, MATLAB, Python, Linux, Git, Docker, AWS (EC2, S3, CloudWatch)

Semiconductor Packaging – Advanced Semiconductor Packages, Soldered Packages, Socketed Packages, Die-Level Behavior, Interconnect Integrity, BEOL Stress, Substrate Cracking, Delamination, HBM Integration, 2.5D Interposers, 3D Packaging, FCBGA, FCLGA/HFCBGA, Chiplet-based Designs

Reliability & Testing – Structural Reliability, Life Prediction Models, Qualification Testing, Manufacturing Data, Material Characterization, Reliability Testing, System-Level Test Correlation

Platform & System Interaction – PCB Interaction, Socket and Retention Systems, Heatsink/Cold Plate Interaction, Thermal Interface Materials (TIM1.5), Platform Engineering, System Integration

PROJECTS

Server SoC Structural Integrity Analysis

- Developed detailed **finite element models** of server SoCs and their associated semiconductor packages to predict structural behavior under thermal and mechanical loads.
- Applied **ANSYS APDL** scripts to automate meshing, boundary condition application, and analysis setup for complex package geometries, reducing manual effort by approximately **30%**.
- Performed sensitivity analyses on critical material properties and interface behaviors, identifying key drivers for **warpage and stress** in socketed configurations.

Package-to-Platform Mechanical Coupling Simulation

- Modeled the mechanical interaction between server SoC packages (both soldered and socketed) and the host Printed Circuit Board (PCB), including socket and retention system effects.
- Integrated **thermal interface material (TIM1.5)** models to assess performance under combined mechanical compression and thermal gradients using coupled physics simulations.
- Validated simulation results against limited experimental data for package deformation, contributing to a more accurate understanding of **system-level mechanical integrity**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)